Radiologia abdominal básica

César Tavaniello Neto

R1 Radiologia e Diagnóstico por Imagem ISCMPA

Porto Alegre, 01 de abril de 2025

Ulster Med J 2013;**82(3):**179-187

Pictorial Review

The Abdominal Radiograph

Barry James, Barry Kelly

Accepted 8 August 2013

EDITOR'S CHOICE

Gas patterns on plain abdominal radiographs: a pictorial review

Rachel E Musson,[1] Ian Bickle,[1] Ram K P Vijay[2]

[1]Royal Hallamshire Hospital, Sheffield Teaching Hospitals NHS Trust, Sheffield, South Yorkshire, UK
[2]Barnsley Hospital NHS Foundation Trust, Barnsley, South Yorkshire, UK

Correspondence to
Dr R E Musson, Department of Radiology, C Floor, Royal Hallamshire Hospital, Glossop Rd, Sheffield S10 2JF, UK; rachel.musson@sth.nhs.uk

Received 1 December 2009
Accepted 7 December 2010
Published Online First 17 January 2011

ABSTRACT

Abdominal radiographs are one of the most commonly performed radiological examinations and have an established role in the assessment of the acute abdomen. The main indication is for suspected bowel obstruction and in conjunction with an erect chest x-ray for suspected visceral perforation. Often, the pattern of gas points to a particular pathology, and accurate interpretation is important for prompt diagnosis. The diagnosis in most cases will be confirmed by further imaging studies such as ultrasound, contrast studies or, most commonly in contemporary practice, CT. This pictorial review summarises the various types of intraluminal and extraluminal gas patterns, illustrates some of the common clinical diagnoses made from plain films, describes some commonly encountered clinical problems with radiographs, and discusses the role of advanced imaging techniques.

specifically on the analysis of patterns of gas. Normal gas in the abdomen is predominantly due to swallowed air.

Air in the stomach is often seen, the pattern being somewhat variable with air fluid levels being commonly present. Small bowel air usually appears as multiple, small, randomly distributed, gaseous foci scattered throughout the abdomen. Small bowel gas is increased in patients who chronically swallow air or drink carbonated beverages. A normal small bowel gas pattern varies from no gas being visible to gas in three or four variably shaped small intestinal loops.

It is usually possible to differentiate between dilated small and large bowel on a plain abdominal radiograph. Clues include the distribution of bowel loops—the small bowel is usually located centrally and the large bowel is usually peripheral. This is

Técnica

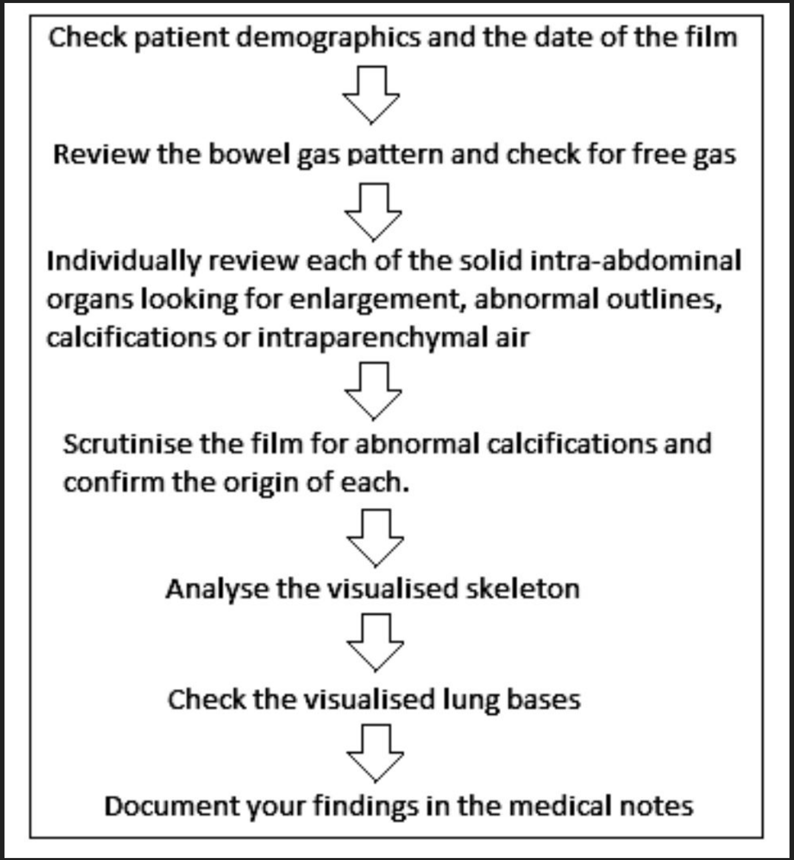
Posição supina, projeção AP

Diafragma até os ramos púbicos inferiores, incluindo a musculatura da parede abdominal lateral

Alta dose de radiação (1,5mSv):

- 75 radiografias de tórax
- Um sexto da dose de uma TC padrão de abdômen

Interpretação



PADRÃO DE GASES INTESTINAIS - Ar livre

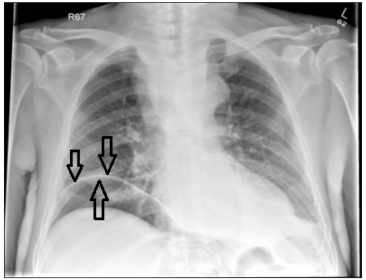


Fig 1a. Erect Chest Radiograph in a patient with acute epigastric pain on a background of alcohol abuse. Note the well demarcated right hemidiaphragm with air density on both sides (black arrows), indicating air is present in the right subphrenic space. A perforated duodenal ulcer was found at laparotomy.

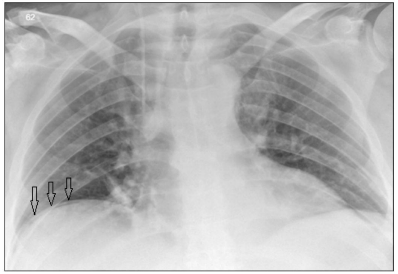


Fig 1b. Erect Chest Radiograph in a patient with acute-on-chronic left iliac fossa pain. There is a tiny focus of air density below the right hemidiaphragm (black arrows) again indicative of free intraperitoneal air. This patient was found to have a perforated sigmoid diverticulum on a subsequent CT scan.

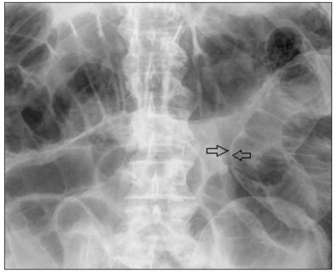


Fig 2. Large bowel obstruction with perforation. There is a distended loop of transverse colon in the upper abdomen (see below). There is a distinctly abnormal bowel loop on the left side of the radiograph. On closer inspection, it is noted that there is air density on both sides of the bowel wall (arrows) indicating a large volume of free intraperitoneal air. This is known as Rigler's sign.

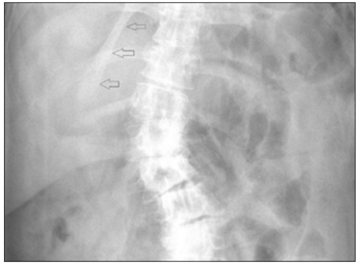


Fig 3. Falciform Ligament Sign. Normally the falciform ligament is surrounded by soft tissue and therefore is not visible on the abdominal radiograph. However, if there is free air in the abdominal cavity, it surrounds the falciform ligament making it visible. In the example above, the falciform ligament is clearly outlined by air (black arrows).

Ar no espaço subfrênico

Sinal de Rigler

Presença de ar em ambos os lados da parede intestinal

Sinal do Ligamento Falciforme

A presença de ar delinea o ligamento falciforme

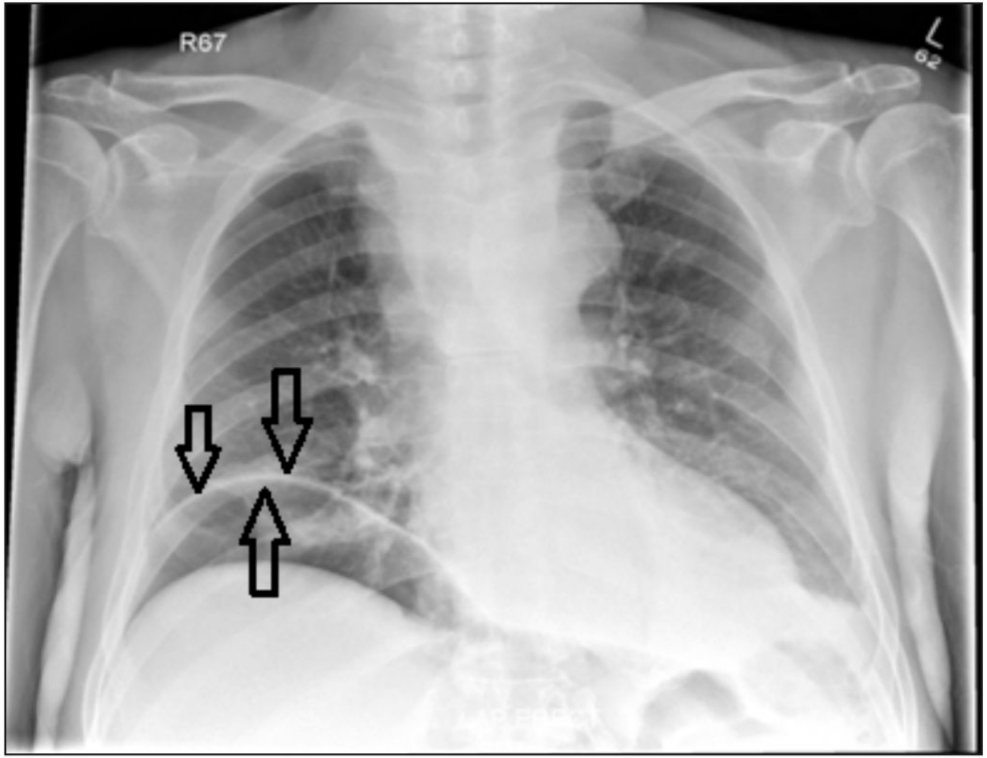


Fig 1a. Erect Chest Radiograph in a patient with acute epigastric pain on a background of alcohol abuse. Note the well demarcated right hemidiaphragm with air density on both sides (black arrows), indicating air is present in the right subphrenic space. A perforated duodenal ulcer was found at laparotomy.

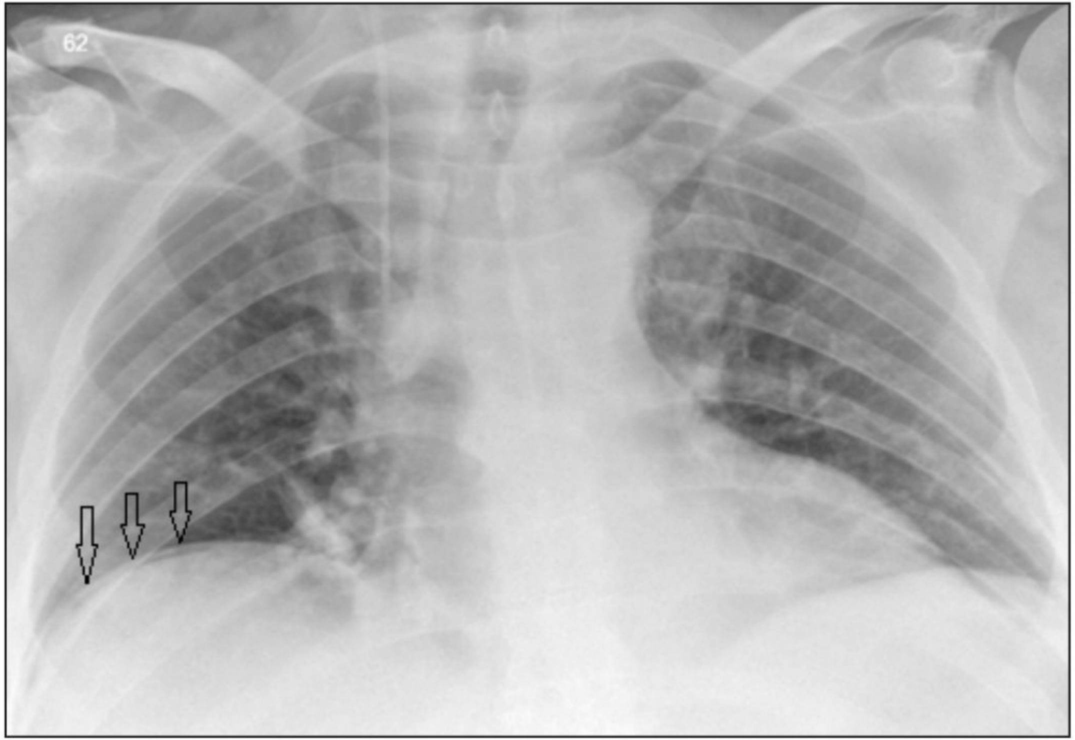


Fig 1b. Erect Chest Radiograph in a patient with acute-on-chronic left iliac fossa pain. There is a tiny focus of air density below the right hemidiaphragm (black arrows) again indicative of free intraperitoneal air. This patient was found to have a perforated sigmoid diverticulum on a subsequent CT scan.

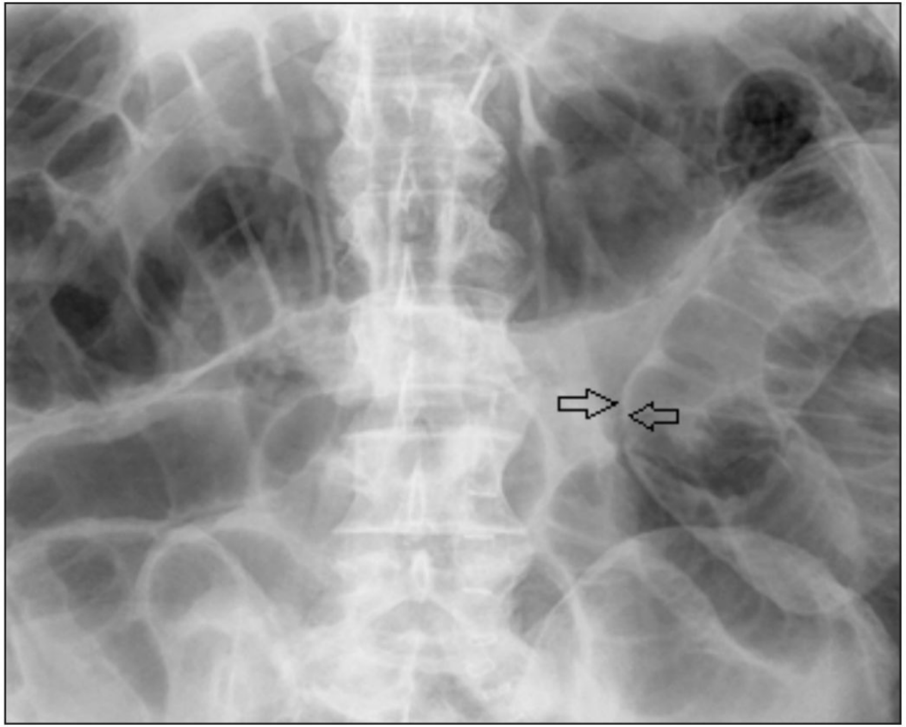


Fig 2. Large bowel obstruction with perforation. There is a distended loop of transverse colon in the upper abdomen (see below). There is a distinctly abnormal bowel loop on the left side of the radiograph. On closer inspection, it is noted that there is air density on both sides of the bowel wall (arrows) indicating a large volume of free intraperitoneal air. This is known as Rigler's sign.

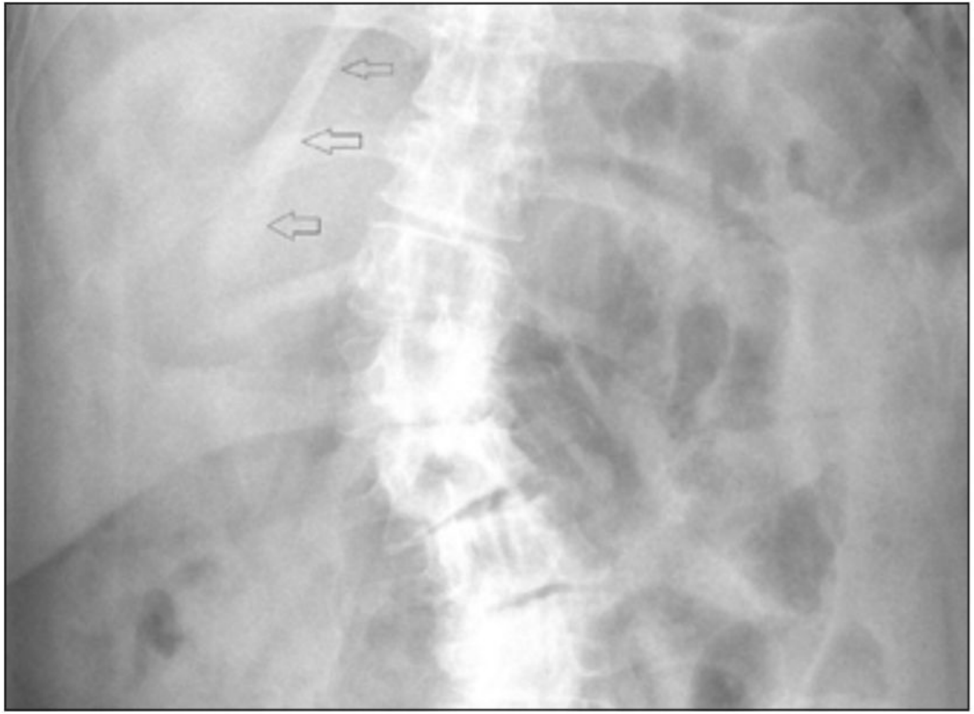


Fig 3. Falciform Ligament Sign. Normally the falciform ligament is surrounded by soft tissue and therefore is not visible on the abdominal radiograph. However, if there is free air in the abdominal cavity, it surrounds the falciform ligament making it visible. In the example above, the falciform ligament is clearly outlined by air (black arrows).

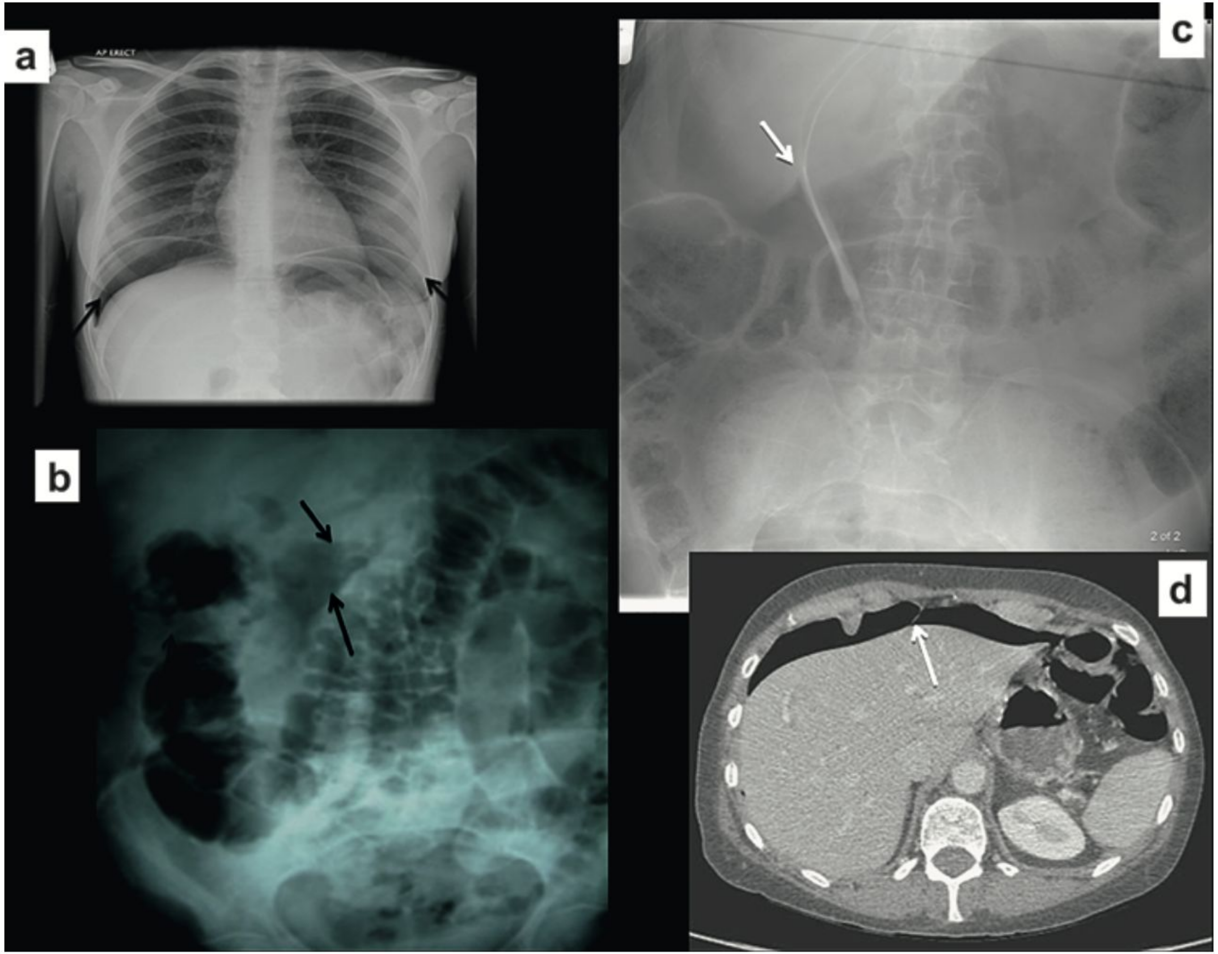


Figure 14 Signs of pneumoperitoneum I. (A) Free sub-diaphragmatic gas on chest radiograph (arrows). (B) Triangle of free peritoneal gas (arrows). (C,D) Abdominal radiograph and complementary CT scan showing the falciform ligament sign (arrows).

"Intraperitoneal free air is a normal finding in the early post-operative period but it should be viewed with suspicion if it persists beyond 6 days or if the patient is clinically unwell."

PADRÃO DE GASES INTESTINAIS - Obstruções

Intestino Delgado

Intestino Grosso

Volvos

Obstrução de Intestino Delgado

Localização central

Válvulas coniventes que atravessam todo o diâmetro do lúmen intestinal

O intestino delgado normal não deve exceder 3 cm de diâmetro

Íleo (generalizado ou localizado) X Obstrução mecânica (75% aderências, hérnias)

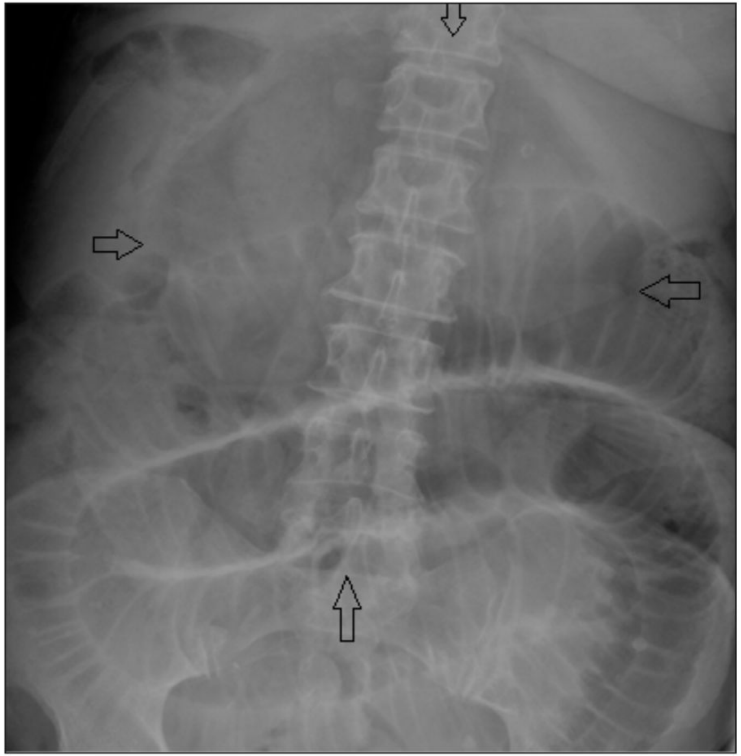


Fig 4. Small bowel obstruction with perforation. Note the dilated loops of bowel centrally within the abdomen. There are mucosal folds spanning the entire width of the bowel wall indicating these represent loops of small bowel. There is an ovoid air density projected over the upper abdomen which is called the "Football Sign" (black arrows) and is in keeping with massive pneumoperitoneum.

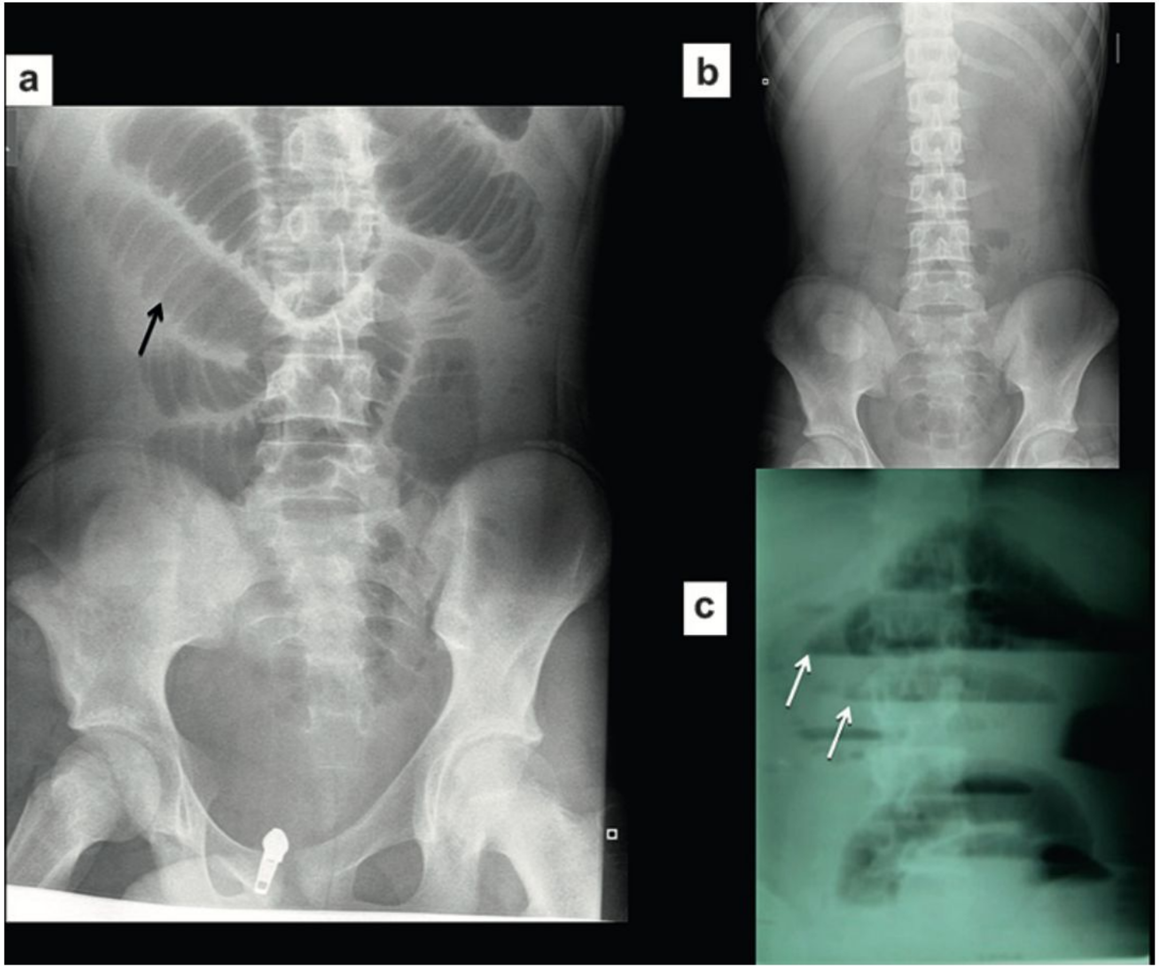


Figure 1 Small bowel obstruction. (A) Multiple loops of gas-filled distended small bowel. Note the valvulae conniventes extending across the bowel circumferentially (arrow). (B) A paucity of abdominal gas due to fluid-filled obstructed small bowel. (C) An erect abdominal radiograph showing multiple fluid levels—'step ladder' pattern (arrows).

Obstrução de Intestino Grosso

Localização periférica

Haustrações

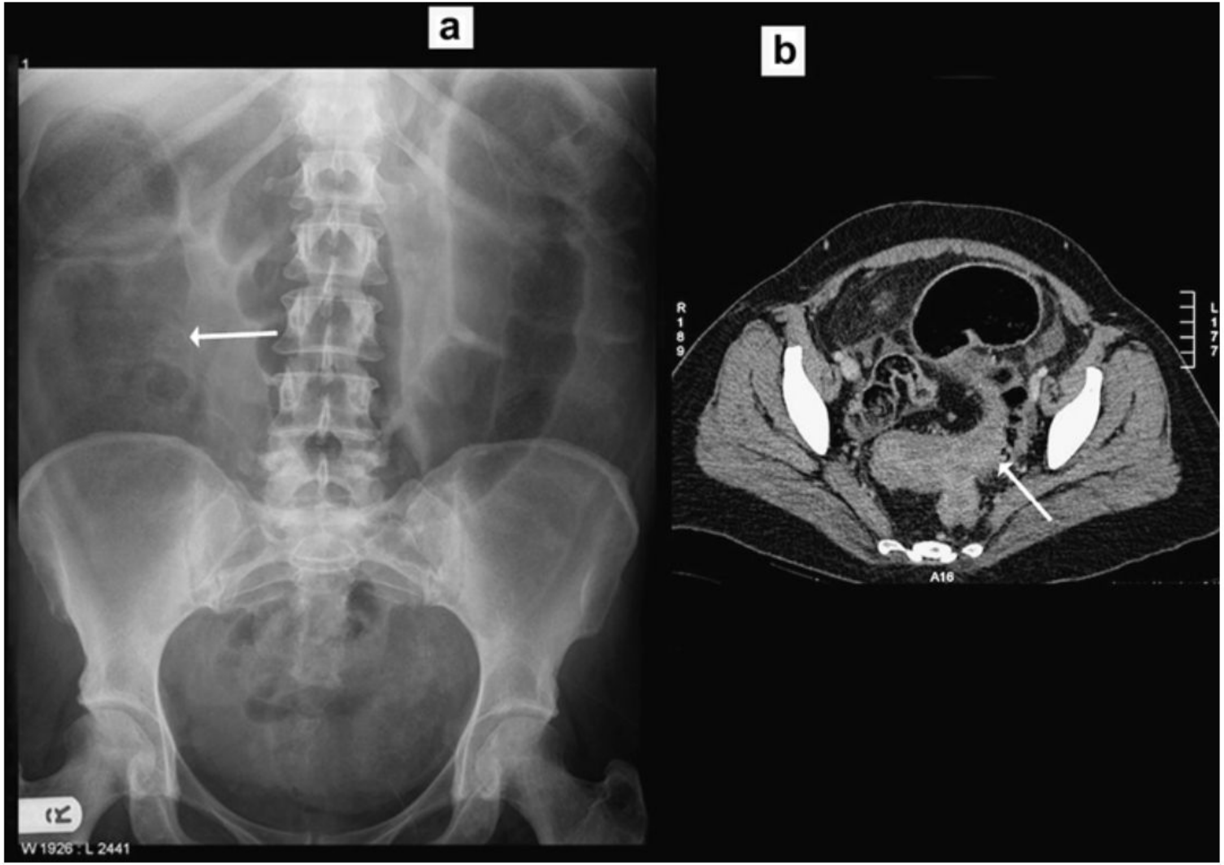
Estruturas fixas e variáveis

Dilatação se cólon >6 cm ou ceco >9 cm

Obstrução em alça fechada e aberta

Fig 5. Large bowel obstruction – open loop. There is a dilated loop of large bowel in the right side of the abdomen – note the mucosal folds do not cross the entire width of the bowel wall. There is also small bowel dilatation centrally within the abdomen indicating that the ileocaecal valve is incompetent. This patient was found to have an obstructing colonic tumour at the hepatic flexure.

Figure 3 Large bowel obstruction. (A) Plain radiograph showing distended large bowel loops. Note grossly dilated caecum (white arrow). (B) Subsequent CT scan showing dilated large bowel with a transition point in the pelvis due to an obstructing carcinoma of the sigmoid colon (white arrow).



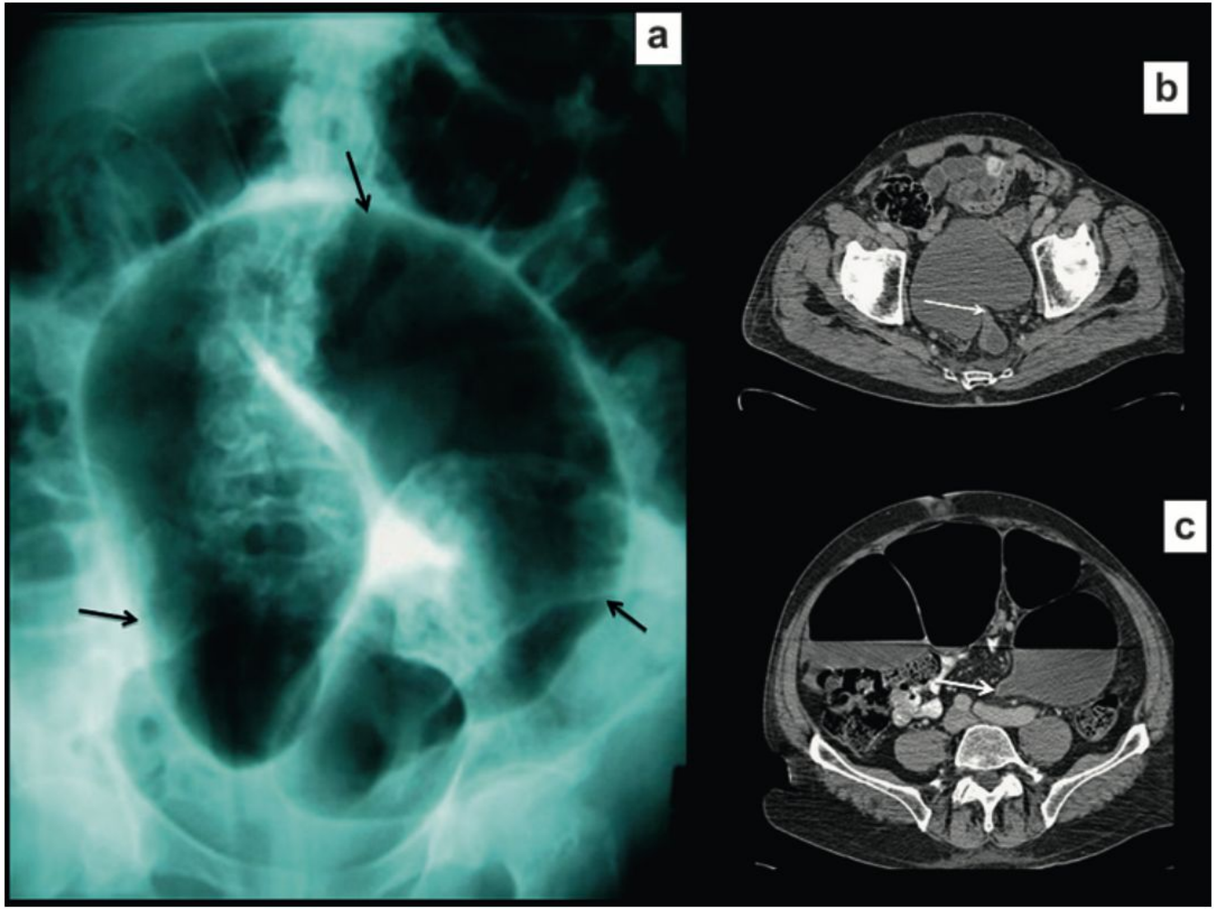
Volvos

Cólon se torce em seu mesentério e forma uma obstrução em alça fechada

Alça em forma de U invertida → a aposição das paredes mediais das alças forma o clássico ¨**Sinal do Grão de Café**¨

Sigmoide é a topografia mais comum, seguida pelo ceco. Esta, dependendo da competência da válvula ileocecal, pode acarretar em obstrução de alça aberta ou fechada

Figure 4 Sigmoid volvulus. (A) Plain film showing a large distended loop of sigmoid colon arising from the pelvis extending up to the right upper quadrant, with proximal large bowel obstruction. This is the typical appearance of sigmoid volvulus (coffee bean sign; arrows). (B,C) A confirmed sigmoid volvulus on the CT scan. Note the ‘bird’s beak’ appearance (white arrows) of the twisted sigmoid colon mesentery.



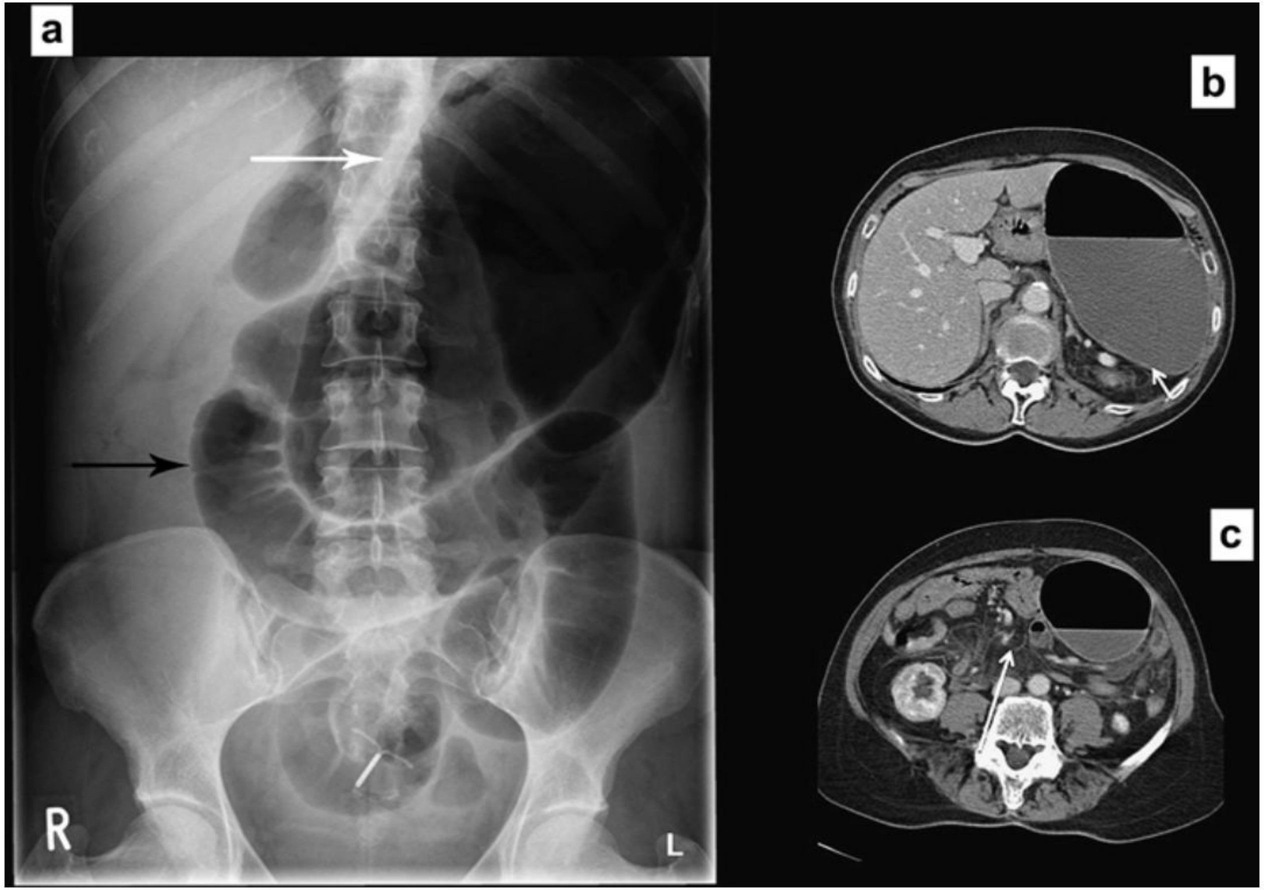


Figure 5 Caecal volvulus. (A) Erect abdominal radiograph showing a grossly dilated gas-filled loop of bowel (caecum) projected in the upper abdomen (white arrow). Note the lack of gas in the rest of the large bowel and dilated small bowel loops (black arrow). (B,C) This was confirmed on the CT scan, which also shows twisted caecal mesentery.

COLITE

Sinal da “impressão do polegar”: indicador de colite, mas não é específico

Dobras mucosas edematosas que parecem impressões digitais ao longo da parede do cólon cheio de gás

Megacólon tóxico

Definido pela dilatação do cólon transverso >6 cm ou pela distensão colônica progressiva em radiografias em série


R39

Colite crônica

Atrofia e cicatrização da mucosa colônica

Cólon em "tubo de chumbo"

Ausência da haustra colônica normal e formação de duas paredes colônicas paralelas sem características

Fig 8a. Colitis. Note the thickened mucosal folds of the left hemicolon, often referred to as "thumbprinting". Fig 8b. The transverse colon is featureless in keeping with chronic colitis.

AR NO FÍGADO

PNEUMOBILIA	GÁS VENOSO PORTAL
Comunicação anormal entre o intestino e a árvore biliar	Outra importante causa de sombra de gás sobre o fígado
Esfincterotomia endoscópica prévia Fístulas colecistoentéricas em casos de íleo biliar (tríade: dilatação intestino delgado + pneumobilia + cálculo biliar fora da vesícula)	Isquemia intestinal Necrose
Gás na região **central** do fígado	Gás na **periferia** do fígado

Diferenciação entre as duas condições é definida por meio de tomografia abdominal

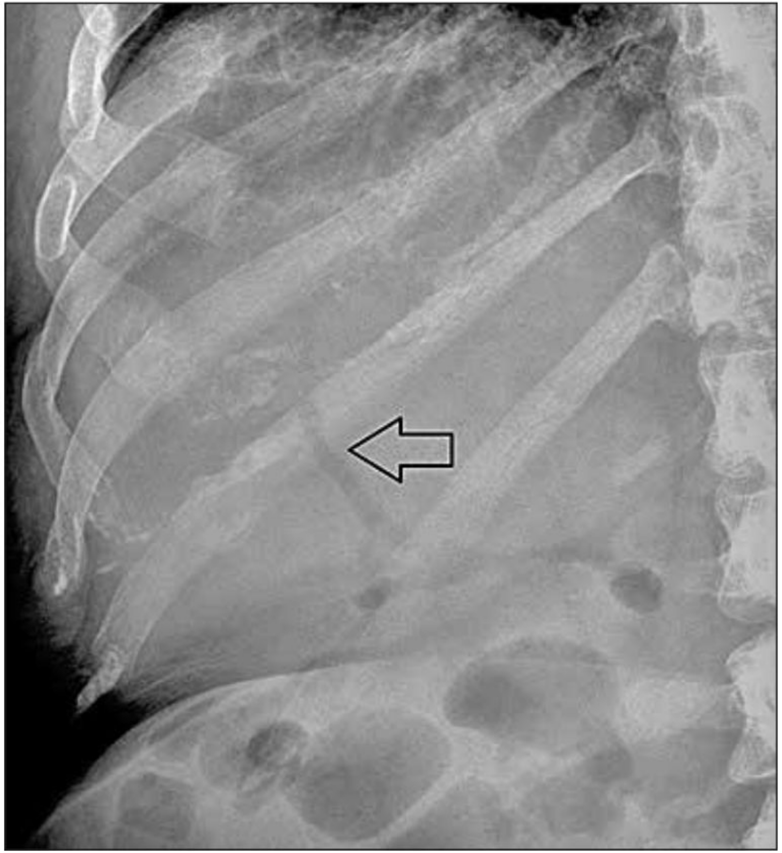


Fig 9a. Pneumobilia. Linear gas shadowing is noted centrally within the liver in the region of the porta hepatis (black arrow). This patient had a previous sphincterotomy.

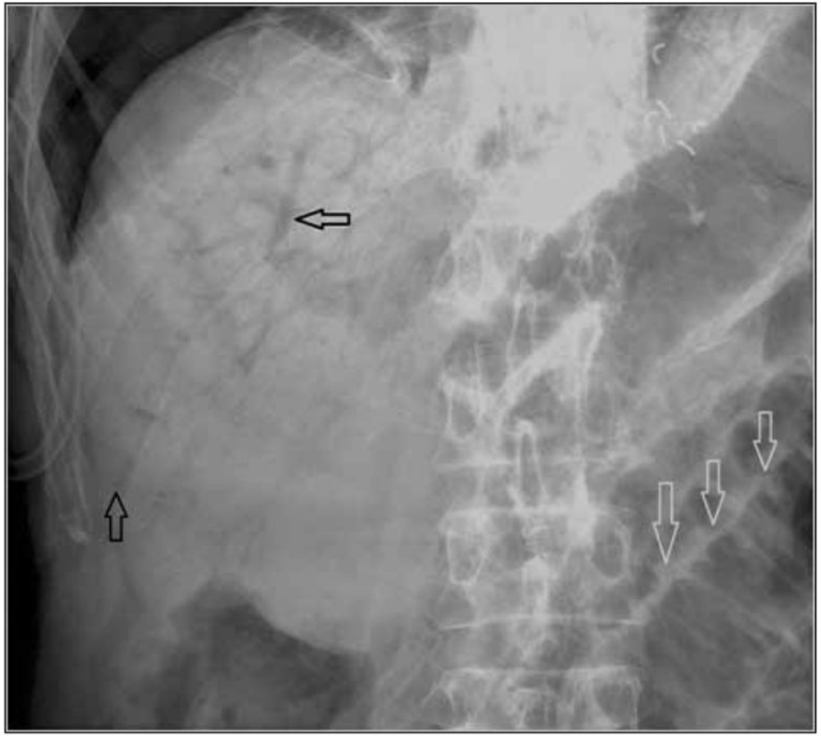


Fig 9b. Portal venous gas. This moribund patient was known to have small bowel ischaemia. The AXR demonstrates extensive linear gas shadowing with characteristic extension to the peripheries of the liver. This pattern is in keeping with portal venous gas, an ominous radiological finding. Distended loops of small bowel are noted in the left lower corner of the film (white arrows).

CALCIFICAÇÕES - cálculos do trato renal

Radiografia KUB está sendo substituída por TC de baixa dose (principal modalidade hoje)

Contorno renal
Pesquisa ativa por cálculos
Trajeto dos ureteres
Junções JUP e JUV
Bexiga

Fig 11. Intravenous Urogram. Contrast is seen to fill the renal pelvicalyceal systems, the ureters and the bladder. This image illustrates the normal course of the ureters through the abdominal cavity.

CALCIFICAÇÕES

Flebólitos: pequenas calcificações murais venosas

Podem ser confundidos com cálculos ureterais distais

Ovoide com centro transparente X Irregulares e uniformemente densos

Apendicite aguda

Apendicolitos: calcificações em fossa ilíaca direita

CALCIFICAÇÕES - cálculos biliares

Ultrassonografia é a modalidade de escolha para confirmar a presença de cálculos na vesícula biliar

Apenas 10% dos cálculos biliares conterão calcificação suficiente para serem visíveis na radiografia abdominal

Fig 10. Staghorn Calculi. Unlike Fig 11, no contrast has been administered to this patient. The dense calcification seen in both collecting systems represents renal calculi (white arrows). Incidental note is made of a gallstone in the right upper quadrant (black arrow). Note the difference in the shape and type of calcification seen in gallstones.

CALCIFICAÇÕES - vasculares

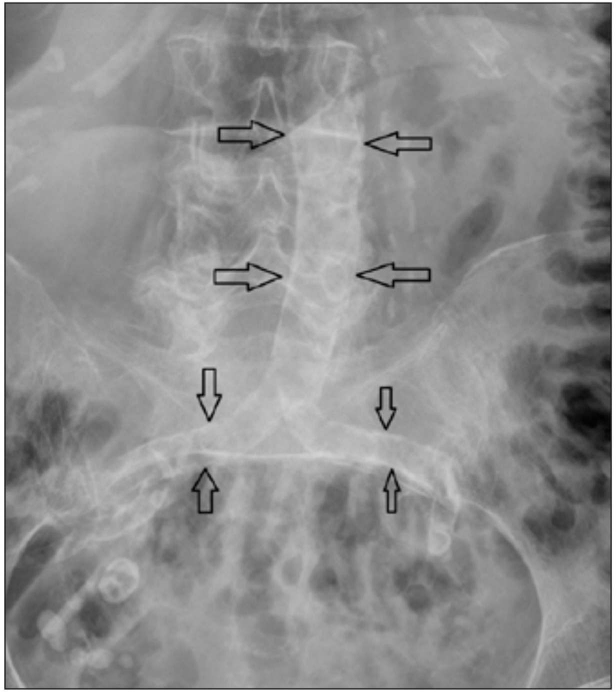


Fig 12. Dense calcified atherosclerotic plaque is noted in the aorta and the common iliac arteries. The aortic walls remain parallel indicating there is no aneurysmal dilatation.

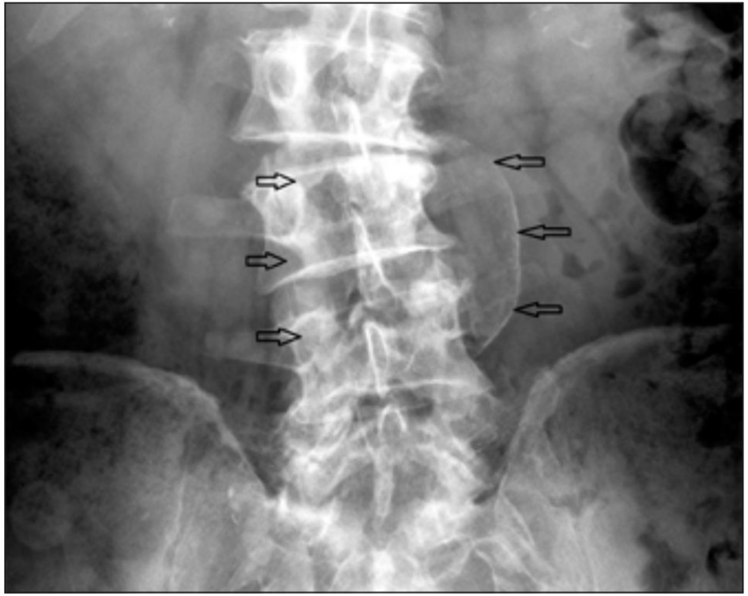


Fig 13. Coned image of an abdominal aortic aneurysm. There is loss of the normal parallel orientation of the calcified aortic walls. The black arrows demonstrate the margins of the aortic sac. Ultrasound or CT will ultimately be required to accurately assess the morphology of the aneurysm and guide treatment or follow-up.

OSSOS

Avaliação das costelas inferiores, coluna lombar, sacro, pelve e fêmur proximal

Lesões ósseas focais

Doenças da coluna e das articulações

Metástases líticas ou escleróticas (topografia de coluna lombar e pelve)

Fraturas osteoporóticas agudas

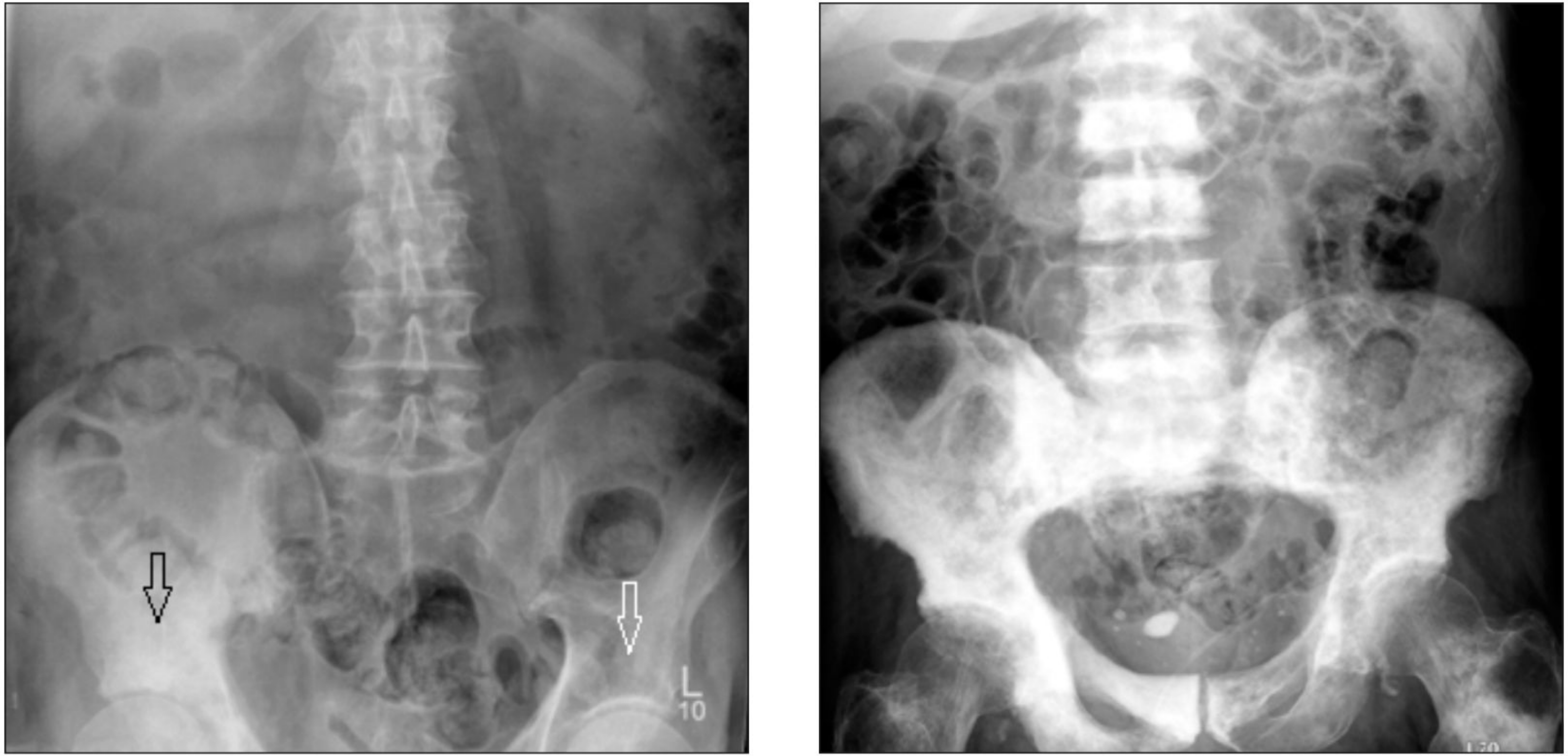


Fig 14a . Pelvic Metastasis. This patient presented with right iliac fossa pain. Note the sclerotic metastasis in the right acetabular roof (black arrow) which is most apparent when comparison is made with the left acetabular roof (white arrow). The patient was later found to have metastatic prostate carcinoma. Fig 14b. Widespread sclerotic metastases are noted throughout the visualised axial skeleton.

OSSOS

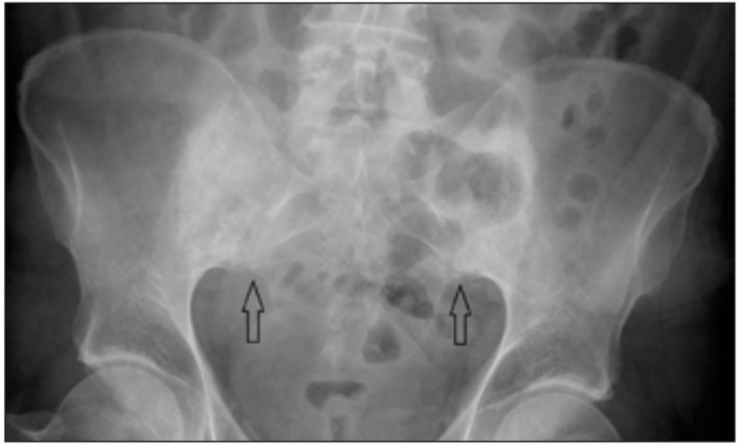


Fig. 15. Bilateral chronic sacroiliitis (black arrows). There is loss of the sacroiliac joint space bilaterally with adjacent subchondral sclerosis.

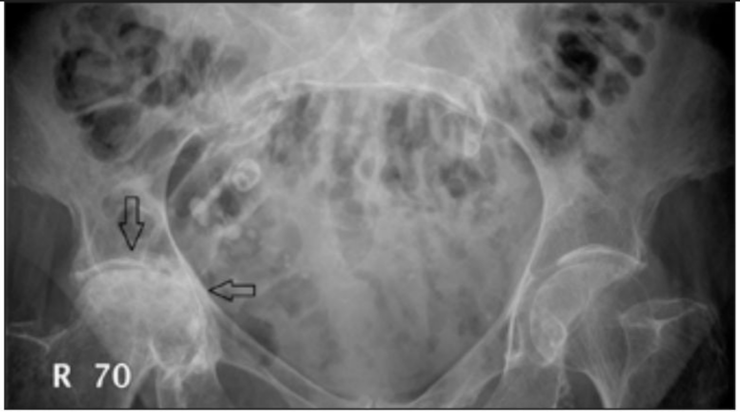


Fig 16. Right hip osteoarthrosis. There are advanced degenerative changes in the central portion of the right hip joint with loss of joint space, subchondral cyst formation and subchondral sclerosis. There is also flattening of the superomedial aspect of the femoral head articular surface.

Sacroileíte: esclerose do osso subcondral de cada lado da articulação sacroilíaca, erosões e eventual anquilose da articulação

Osteoartrose: estreitamento do espaço articular, esclerose subcondral/formação de cisto e osteofitose marginal

BASES PULMONARES

Pneumonia basal causando dor epigástrica

Metástases em bases pulmonares